# 16*100/1000M SFP + 8*1G SFP/RJ45 Combo + 4*1G/2.5G/10G SFP+

# Layer 3 Managed Ethernet Switch

# LT-S8528XG-4XS16GX8GC

## Overview

SmartByte LT-S8528XG-4XS16GX8GC is a Layer 3 Managed Gigabit Switch that provides high-density performance, Layer 3 static and dynamic routing, RIP (Routing Information Protocol) and OSPF (Open Shortest Path First). With 128 Gbps switching capacity and 4 *1G/2.5G/10Gbps flexible and higher performance uplink capability. The LT-S8528XG-4XS16GX8GC provides comprehensive end to end QoS as well as flexible and rich management, security settings, which can meet the high speed, secure and smart requirement for small and medium-size enterprise networks based on cost effective price. The powerful network security features also make the LT-S8528XG-4XS16GX8GC perform effective data traffic control for ISP and enterprise VoIP, video streaming, and multicast applications.

## Picture



## Features

- 4*1G/2.5G/10G SFP+ slots uplink, 16*100/1000M SFP ports + 8*1G SFP/RJ45 COMBO ports downlink, 1*RJ45 console interface for switch basic management and setup.
- **Hardware:** Marvell AC5 series 1.2GHz CPU, Marvell AC5 series switch chipsets, 1GB RAM and 512MB flash
- **Layer 3 features:** IPv4/IPv6 static routing, DHCPv4/ DHCPv6 client, DHCPv4 relay, DHCP relay; Supports ECMP, VRRP, RIP, OSPF, BGP4, IS-IS; 13K routing entries
- **Port features:** Supports port aggregation, IEEE802.3x flow control, port isolation, port mirroring, loop detection and broadcast storm suppression
- **VLAN:** Supports IEEE802.1Q tagged VLAN, mac based VLAN, IP based VLAN, protocol based VLAN, voice VLAN and private VLAN.
- **QinQ:** Supports port-based, VLAN-based and flow-based QinQ
- Supports **GVRP/LLDP/SPAN/RSPAN/DHCPv4 snooping**
- **Layer 2 ring network protocol:** Supports STP, RSTP, MSTP, G.8032 ERPS protocol, single ring, sub ring
- **Multicast:** IGMP snooping, IGMP v1, v2, v3, PIM-SM
- **QoS:** QoS class, remarking, Supports SP, WRR queue scheduling, Ingress/Egress port-based rate-limit and policy-based QoS
- **Security:** Supports ACL, 802.1x， port-security, IP source guard, ARP-check, DAI, anti-gateway ARP spoofing, Ddos attack protection, telnet/SSH/HTTP/HTTPS user access control and AAA.
- **Reliability:** Loop detect based on port or VLAN, the Ethernet in the first mile 802.3ah
- **Management:** Supports IPv4/IPv6 dual stack management, system log management, user management, login service management, firmware management, web management, CLI management, fault detection, DNS, NTP, sFlow, SNMP, MQTT, RMON and dual partition management.
- **Application protocol:** Supports telnet server, telnet client, SCP, SFTP, SSH server, TACACS and TFTP client

## Technical Specification

| | | | |
|---|---|---|---|
| **Model No.** | LT-S8528XG-4XS16GX8GC | | |
| **Interface** | 1G/2.5G/10G Fiber ports | 100/1000 Fiber ports | Gigabit Combo |
| | 4 | 16 | 8 |
| **Ethernet** | 16*100/1000Base-X SFP<br>8*COMBO SFP/RJ45<br>4*1G/2.5G/10GBase-X SFP+ | | |
| **CPU** | Marvell AC5 series 1.2GHz | | |
| **Size of RAM** | 1GB | | |
| **Size of Flash** | 512MB | | |
| **Switch chip model** | Marvell AC5 series | | |
| **Management port** | 1*RJ45 Console port (115200,8,N,1), 1*USB 2.0 | | |
| **Standard** | IEEE 802.3 10Base-T Ethernet<br>IEEE 802.3u 100Base-TX Ethernet<br>IEEE 802.3ab 1000Base-T Ethernet<br>IEEE 802.3z 1000Base-X Ethernet<br>IEEE 802.3ae 10Gigabit Ethernet<br>IEEE802.3x Flow Control and Back Pressure<br>IEEE 802.1D Spanning Tree Protocol<br>IEEE 802.1w Rapid Spanning Tree Protocol<br>IEEE 802.1Q VLAN<br>ITU-T G.8032 ERPS<br>IEEE 802.1X Port Authentication Network Control<br>IEEE 802.1ab LLDP<br>IEEE 802.3ad LACP | | |
| **LED Indicators** | | | |
| **PWR** (Power indicator) | Off: the device is power off or failed | On: the device power on is normal | |
| **SYS** (System status indicator) | Blinking: device initialization | On: device on normal operation | |
| **1-8** (Copper ports indicators) Green | Off: ports link down | | |
| | On: ports link up | | |
| | Blinking: data on TX/RX | | |
| **1-8** (Copper speed indicators) Yellow | Off: ports on 10/100M | | |
| | On: ports on 1000M | | |
| **1-8**(Fiber ports indicators) | Off: ports link down | | |

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| Green | On: ports link up |
| | Blinking: data on TX/RX |
| **9-28**(Fiber ports indicators) Green | Off: ports link down |
| | On: ports link up |
| | Blinking: data on TX/RX |
| **Power parameter** | |
| Input voltage | 1* AC 100-240V input **on default**<br>1* DC 36-72V input optional<br>2* AC 100-240V input optional<br>1* AC 100-240V input + 1* DC 36-72V input optional |
| Total power consumption | Full loading ≤36W |
| Over-current protection | Support |
| **Layer 2 switching** | |
| Port aggregation | Support GE port aggregation |
| | Support 2.5GE aggregation |
| | Support 10GE aggregation |
| | Support static aggregation |
| | Support LACP dynamic aggregation |
| | Up to 64 aggregation groups and up to 8 ports per group |
| Port features | Support IEEE802.3x flow control |
| | Support interface counters |
| | Support port isolation |
| | Port mirroring (One-to-One, Many-to-One) |
| | Support loop detection (Port-based; VLAN-based) |
| | Support broadcast storm suppression (broadcast; unknown multicast; unknown unicast) |
| VLAN | IEEE802.1Q tagged VLAN |
| | Mac based VLAN |
| | IP based VLAN |
| | Protocol based VLAN |
| | Voice VLAN, specially divide VLAN for user voice data flow |

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| | Private VLAN, divide a VLAN broadcast domain into multiple subdomains to form a double-layer VLAN structure |
| | VLAN range: 1-4094 |
| | Support access mode/trunk mode/hybrid mode |
| Mac-address table management | Support static Mac-address management |
| | Support dynamic Mac-address Management |
| | Support filtering Mac-address |
| | Support MAC limit based on port and VLAN |
| | Support MAC flapping based on port and VLAN |
| | 16K Mac-address table size |
| GVRP | Normal mode |
| | Fixed mode |
| | Forbidden mode |
| QinQ | Port-based QinQ |
| | VLAN-based QinQ |
| | VLAN mapping |
| | Flow-based QinQ |
| LLDP | Link Layer Discovery Protocol |
| | Med |
| SPAN | Number of Sessions: 7 |
| | Many-to-one |
| RSPAN | Remote Switch Port Analyzer |
| MSTP | 802.1D STP |
| | 802.1W RSTP |
| | 802.1S MSTP |
| | BPDU Guard |
| | Number of Single Ring Device Connections: 20 |
| ERPS G.8032 | Major ring |
| | Sub ring |
| | Recovery time ≤20ms |

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| | Number of Single Ring Device Connections: No limit |
| QoS | QoS Class, Remarking |
| | Support SP, WRR Queue Scheduling |
| | Ingress Port-based Rate-limit |
| | Egress Port-based Rate-limit |
| | Egress port-queue shaper |
| | Policy-based QoS |
| | Port Queue Numbers: 8 |
| DHCPv4 Snooping | DHCP snooping trust port |
| | DHCP snooping option-82 |
| | DHCP snooping circuit-id and remote-id customized |
| **L2 Multicast** | |
| IGMP | IGMP v1, v2, v3 |
| IGMP Snooping | Internet Group Management Protocol Snooping |
| | Fast Leave |
| | Multicast Table Size: 1K |
| PIM-SM | Protocol Independent Multicast-Sparse Mode |
| **Security features** | |
| ACL | IP Standard ACL<br>IP Extended ACL<br>MAC extend ACL<br>IPv6 ACL<br>Time Range<br>ACL apply on VLAN<br>ACL entries, include input and output |
| 802.1x | Port access control<br>Mac-address access control<br>RADIUS server |
| Port-security | Port-security |
| IP source guard | IP Source Guard, Port-based IP/Mac Binding Access |
| ARP-check | Support ARP-check and ARP packet filtering for illegal users |
| DAI | Dynamic ARP Inspection |
| Anti-gateway ARP Spoofing | Prevent Gateway ARP Spoofing |

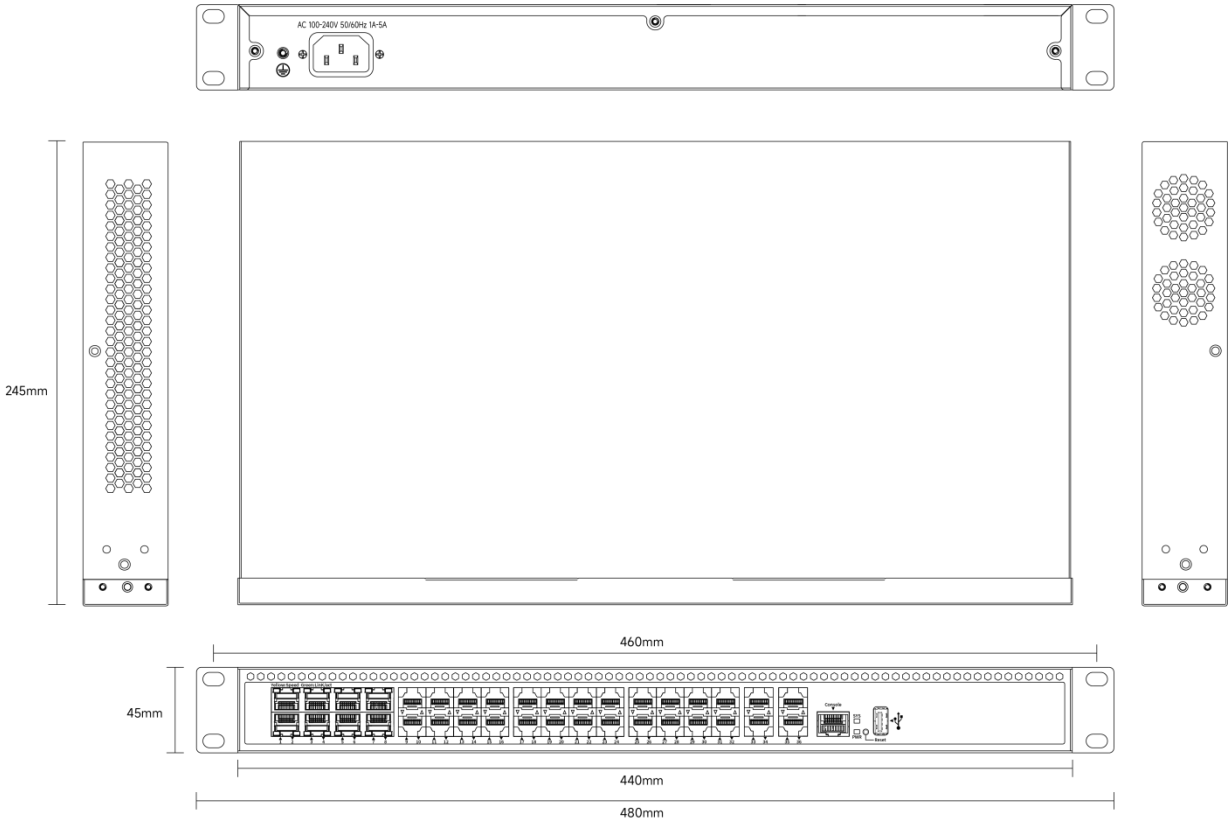
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| Ddos | Ddos attack protection |
| Access control | Support Telnet/SSH/HTTP/HTTPS user access control |
| AAA | RADIUS authentication access switch |
| | TACACS authentication access switch |
| | Authorization |
| **Reliability** | |
| Loop Detect | Loop detect, based on port or VLAN |
| EFM | The Ethernet in the first mile, 802.3ah |
| **Management & Maintenance functions** | |
| Dual Stack | IPv4 and IPv6 Dual Stack Management |
| System Log Management | Syslog Level 7 to 0<br>Corresponding Debug/Info/Notice/Warning/ Err/ Critical/ Alert |
| | Support connecting up to 3 Servers |
| | For IPv4 and IPv6 |
| User Management | Support password protection, support MD5/SHA256/SHA512 algorithm |
| | Local Management of User Privilege |
| | Remote authorization by RADIUS or TARCAS |
| | Enable privilege protection |
| | User password complexity requirements |
| | User login fail protection |
| | User Privilege Level 0 to 15 |
| Login Service Management | Up to 8 users online simultaneously |
| | Access control based on IPv4 or IPv6 ACL |
| Firmware Management | Firmware upgraded by WEB |
| | Firmware upgraded by CLI |
| | Dual partition |
| Web Management | Web Management |
| | Support HTTP V1.1 |
| | Support HTTPS |
| | For IPv4 and IPv6 |

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| CLI management | Console/Telnet command line management |
| Fault Detection | Ping/Traceroute, For IPv4 and IPv6 |
| | Dying gasp optional |
| | Optical transceiver DDM |
| | One click fault information collection |
| | Cable detection |
| DNS | Domain Name System |
| NTP | Network Time Protocol, For IPv4 and IPv6 |
| | Support domain name configuration |
| Configuration Import /Export | FTP/TFTP based remote configuration import and export |
| sFlow | Sampled flow |
| SNMP | Version SNMPV2C/V3 |
| MQTT | Message Queuing Telemetry Transport |
| RMON | Support RMON (Remote Monitoring) alarm |
| Dual partition management | Support dual partition switching |
| **Application Protocol Functions** | |
| Telnet server | Support telnet server for IPv4 and IPv6 |
| Telnet client | Support telnet client for IPv4 and IPv6 |
| SCP (Secure Copy Protocol) | SCP client/server for IPv4 and IPv6 |
| SFTP | sftp client/server for IPv4 and IPv6 |
| SSH server | Support login based on user password-SSH2, support IPv4, IPv6 |
| | Support public key-based login |
| TACACS | Support TACACS (the terminal access controller accesses the control system) |
| TFTP | Support TFTP Client |
| **Layer 3 features** | |
| Static Route | Support VLAN interface and route port |
| | IPv4 static routing table entry configuration |
| | ARP dynamic learning and static configuration |
| | IPv6 static routing table entry configuration |
| | Neighbor discovery and static configuration |

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| | Routing prefix table size 4k for IPv4, for IPv6 divide by 4 |
| DHCPv4 Client | Dynamic host configuration protocol for IPv4 |
| DHCPv6 Client | Dynamic host configuration protocol for IPv6 |
| DHCPv4 Relay | DHCP Relay for IPv4 |
| DHCPv4 Server | DHCP Server for IPv4 |
| ECMP | Support ECMP (Equal-cost multi-path routing) |
| | Support the configuration of ECMP Max next-hop |
| | Support capacity balanced configuration |
| Route policy | IPv4 prefix-list |
| VRRP | Virtual Router Redundancy Protocol |
| **Routing Protocol Specification** | |
| RIP | RIPv1/v2 |
| OSPFv2 | OSPFv2 |
| BGP4 | BGP4 |
| | BGP Support Routing Recursive ECMP |
| | Support to view the number of neighbors and up/down state |
| IS-IS | IS-ISv4 |
| Routing entries | 13K |
| **Switching features** | |
| Switching capacity | 128Gbps |
| Packet forwarding rate | 190.4Mpps |
| MAC address table | 16K |
| VLAN | 4K |
| Buffer | 12Mbit |
| Forwarding delay | <5us |
| Jumbo Frame | Support 10Kbytes |
| MDI/MDIX | Support |
| Watchdog | Support |
| **Mechanical structure** | |
| Installation method | Desktop; Rack-mount |

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| Dimension(W*D*H)mm | 440*245*44mm |
| Weight | 3.65kg |
| **Operating environment** | |
| Operating temperature | 0℃~ +50℃ |
| Storage/transportation temperature | -10℃~ +70℃ |
| Relative humidity | Operation humidity: 10%-90%RH<br>Storage humidity: 5%-95%RH |
| **Protection** | Surge protection of power: 4KV |
| | Surge protection of Ethernet ports: 4KV |
| | ESD: Contact 6kV, Air 8kV |
| **Certification** | CCC/CE/UKCA/RoHS/REACH/FCC |
| **MTBF** | > 100,000 hours |
| **Warranty** | 3 years |

## Structure diagram



## Order information

| Model | Description |
|---|---|
| LT-S8528XG-4XS16GX8GC | Layer 3 managed Ethernet switch with 4*1G/2.5G/10G SFP+ ports + 16*100/1000M SFP ports，8*1G SFP/RJ45 COMBO ports，AC 100-240V 50/60Hz |
| LT-S8528XG-4XS16GX8GC-2AC | Layer 3 managed Ethernet switch with 4*1G/2.5G/10G SFP+ ports + 16*100/1000M SFP ports，8*1G SFP/RJ45 COMBO ports，dual AC 100-240V 50/60Hz |
| LT-S8528XG-4XS16GX8GC-AC-DC | Layer 3 managed Ethernet switch with 4*1G/2.5G/10G SFP+ ports + 16*100/1000M SFP ports，8*1G SFP/RJ45 COMBO ports，one AC 100 ~ 240V 50/60Hz and one DC 36-72V input |